Deformation Theory Seminar 5
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What does the Hilbert scheme tell us? ~Motivation & Future Guideline~

[Ref] [Har 1]: Deformation theory
[Har 2]: Algebraic geometry
Others: See last page

$\boxed{\text{Deformation theory}}$ = the local study of deformations
≤ "Moduli" if exists.

= infinitesimal study of a "family" in the neighborhood of a given pt.

A model for every situation is Hilb. sch.

Thm 1.1 [Har 1] (Grothendieck, Laudal, Mori)

Y: closed subsch of $X = \mathbb{P}^n_k$ over k: field.

a). $\exists H$: proj sch (called Hilbert sch.) parametrizing closed subsch of X w/ the same Hilb poly. as Y.

- $\exists W \subseteq X \times H$: univ subsch., flat over H s.t. the fibers of W over closed pt $h \in H$ are all closed subschemes of X w/ the same Hilb poly. as Y.

$y \in H \longleftrightarrow Y' \subseteq X$ w/ $p(Y') = P$

$\{W_h\}_{h\in H} \longleftrightarrow \{Y' \subseteq X \text{ w/ } p(Y') = P\}$ "tautological"

- Futher more H is universal i.e. if $^\forall T$: other sch. and if $W' \subseteq X \times T$ has same prop.
$\Rightarrow \exists ! \varphi: T \to H$ s.t. $W' = W \times_H T$ as subsch of $X \times T$

b) $T_y H = H^0(Y, N_{Y/X})$

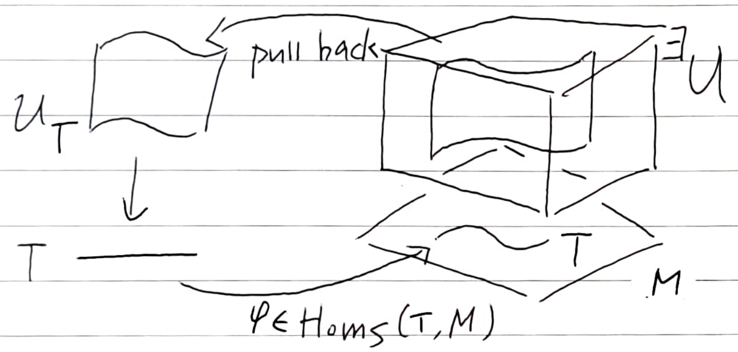
c) If Y: loc. comp. int. & $H^1(Y, N) = 0$
$\Rightarrow H$ is smooth at $y = [Y]$ and $\dim_y H = h^0(Y, N)$

d) In any case, if Y: loc. comp. int.
$\Rightarrow \dim_y H \geq h^0(Y, N) - h^1(Y, N)$

☆ what is "moduli" problem?

Moduli $\overline{\overline{\text{scheme theoretic}}}$ representable functor (by Grothendieck)

$F : (Sch/S)^{op} \to (Sets)$ contravariant
is representable if $\exists M$: sch (called fine moduli sp.)
$F \simeq Hom(-, M)$.



⚠ Often fine moduli sp doesn't exist
→ Coarse modul sp (similar sch but $\nexists$ univ family $\mathcal{U}$)

Hilb.sch suggests two main achievement of deformation theory

- a) ⇒ "infinitesimal ⇒ global"
the existense of a versal deformation ("formal moduli")
containing essentially all informations about all possible deformations
→ See §15 ~ key words: pro-represent
Schlessinger's criterion.

- b), c), d) ⇒ "local study of moduli"
The theory of $\begin{cases} \text{infinitesimal deformations} \\ \text{obstructions.} \end{cases}$
We reduce many geom. problems to cohomological problems.

① Infinitesimal deformation = family over local Artinian ring

$\cup$

which encode "fat" point. i.e
$S = \mathrm{Spec} A$ is the underlying topo. sp $= \{pt\}$
since $S_{red} = \mathrm{Spec} k$
but interesting algebras are functions.

② First order deformation = family over dual number $D = k[\varepsilon]/(\varepsilon^2)$

= Zarisk tangent space of moduli sp if exists

[Har 2] Ex II 2.8.

$\boxed{\begin{array}{l} \text{Let } p : \mathrm{Spec} k \longrightarrow X \quad \text{rational pt} \\ T_p X \cong \{ f : \mathrm{Spec} D \to X \ \text{ s.t } f|_{\mathrm{spec}(k)} = p \} \end{array}}$

pf question is local, so we write
$X = \mathrm{Spec} R$, $p \rightsquigarrow m_p \subset R$: maximal w/ $R/m_p \cong k$.
then $RHS \cong \{ f : R \longrightarrow D : f^{-1}((\varepsilon)) = m_p \}$
only one prime

$\cong \{ f : R/m_p^2 \longrightarrow D : f^{-1}((\varepsilon)) = m_p \}$ $f(m) \subset (\varepsilon)$
since $\varepsilon^2 = 0$, we can define
$\cong \mathrm{Hom}(m_p/m_p^2, k) = T_p X$ $\psi : m \longrightarrow k$
$a \mapsto f(a)/\varepsilon$.

③ Obstruction : $\xi_A \in \mathrm{Def}(A) \rightsquigarrow B \twoheadrightarrow A$ $\exists ?\ \xi_B \in \mathrm{Def}(B)$
We find V: vect sp s.t
for any $\varphi : B \twoheadrightarrow A$, $\exists \vartheta_\varphi : \mathrm{Def}(A) \longrightarrow V$
$\vartheta_\varphi = 0 \Leftrightarrow$ def admits ext. over B
$(V, \{\vartheta_\varphi\})$ tells us when there is an obst to the lifting
= measure of how singular is moduli at the point.
[dimension estimation] $\dim_{\mathbb{C}} T_x M \geq \dim_{\mathbb{C}} M_x \geq \dim_{\mathbb{C}} T_x M - \dim_{\mathbb{C}} V$.

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<u>example of Hilb.sch.</u> (Har 1 Ex 1.1)

$X = V(f) \subseteq \mathbb{P}^2$ curve

f : homogeneous of deg d $\in k[x, y, z] =: S$.

S has $\binom{2+d}{d}$ monomials, we give some order and $N := \binom{2+d}{d} - 1$

$\Rightarrow f = \sum_{i=0}^{N} c_i x^{\alpha_i}$ $\qquad x^{\alpha_i}$: i-th monomial

$f \longleftrightarrow [c_0 : c_1 : \cdots : c_n] \in \mathbb{P}^N = \mathbb{P}(S_d)$

up to scalar.

Now we consider $F = \sum_{i=0}^{N} a_i x^{\alpha_i}$ $\quad a_i$: coordinate on $\mathbb{P}^N$

- $\mathcal{U}_p^2 = V(F)$: tautological family
- $\mathrm{Hilb}_p^2 = \mathbb{P}^N$: Hilb. sch w/ $p(t) = \binom{t+2}{2} - \binom{t+2-d}{2}$ see [Har2] Prop 1.7.6

$$= dt - \frac{d(d-3)}{2}$$

In particular, $d = 2$ (i.e $N = 5$, $p(t) = 2t+1$)

$X = V(y^2 - xz)$: image of Veronese emb $\mathbb{P}^1 \to \mathbb{P}^2$

$X \subseteq \mathbb{P}^2$: Cartier div, then

$N_{X/\mathbb{P}^2} = \mathcal{O}_{\mathbb{P}^2}(X)|_X = \mathcal{O}_{\mathbb{P}^2}(2)|_X = \mathcal{O}_X(2)$ see [Har2] Prop 8.20

this is pull back to $\mathcal{O}_{\mathbb{P}^1}(4)$

$\Rightarrow T_{[X]} \mathrm{Hilb}_p^2 = H^0(X, \mathcal{O}_X(2)) = H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(4)) \cong k^5$

\#

Ⓜ Hilb. sch is best situation, but they are still complicated.

e.g 1 : twisted cubic curve in $\mathbb{P}^3$ (see Har 1 Ex 1.6)

$\Rightarrow$ Hilb sch can be reducible

e.g 2 : Mumford's example (see §13)

$\Rightarrow$ irreducible but generically non-reduced.

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☆ Deformation of OO (See Index of Har 1)

① subsch of a fixed sch (embedded def)

② abstract sch.

③ (local) def of singularity (= affine scheme)

④ quotient

⑤ coherent sheaf

⑥ Invertible sheaf

⑦ Hilb flag sch : see ex 6.8 & Thm 24.7

⑧ morphism : see lem 24.8 ~ ⟹ "bent & break"

⑨ pair (X, L) X: non-singular, L: line bundle : see ex 10.6.

⑩ Cone : see Thm 5.4, Thm 29.11

Appendix Hilb. poly and flatness.

⑪ Hilb poly is $\exists$ poly $h(t)$ s.t

$$h(n) = \chi(\mathbb{P}^n, \mathcal{F}(n)) \ \forall n \in \mathbb{N} \text{ for } \mathcal{F} \text{ over } \mathbb{P}^n$$

In general, $X \subseteq \mathbb{P}^n$: proj var. w/ $\mathcal{L}$: ample

$\Rightarrow h(n) = \chi(X, \mathcal{F} \otimes \mathcal{L}^{\otimes n}) \ \forall n \in \mathbb{N}$

In particular, $\mathcal{F} = \mathcal{O}_X \Rightarrow$ Hilb poly of sch.

⑪ [Har 2] Thm 9.9 T : int. noeth. sch, $X \subseteq \mathbb{P}^n \times T$ closed

$\boxed{\begin{array}{l} \forall t \in T \ P_t := h(X_t) \\ X \text{ is flat over } T \Leftrightarrow P_t \text{ is independent of } t \end{array}}$

① Embedded Deformation

$Y \subseteq X$ closed sch.

Deformation Y' over A

- $Y' \subseteq X' = X \times A$ flat over A

s.t $Y' \times_A k = Y$.

② Abstract Deformation

Y : sch

Deformation (Y', i) over A

- $i : Y \hookrightarrow Y'$ closed imm.

s.t

$$\begin{tikzcd} Y \arrow[r, hook, "i"'] & Y' \arrow[r] \arrow[d] & Y' \times_A k \text{ isom} \arrow[d] \\ & A \arrow[r] & \operatorname{Spec} k \end{tikzcd}$$

with fixed isom with central fiber

why we need i?

$\boxed{\text{with } i}$ $\longleftarrow$ $\boxed{\text{without specifying}}$

/ $\operatorname{Aut}(Y)$ less behaved functorially

infinitesimal automorphism $= \operatorname{Aut}(Y_0 \times_k D / Y_0)$

= "restricting to the identity automorphism on Y_0."

Fact (see Cor 18.3, Rem 10.1.1, Rem 10.2.2)

Sheaf of infinitesimal Aut of sm. var $= H^0(Y, T_Y)$

(This is the analog of the fact in diff. geom. vector fields which are the direction for each point to flow along)

$\operatorname{Aut}(Y)$ mess up the situation and obstruct the existense of moduli space

			Non-sing : Y	③ Affine = pure alg. prob.	
Aut	: Nothing		Aut : $H^0(Y, T_Y)$	Aut : T^0	Rem 10.1.1
1st order	: $H^0(Y, N_{Y/X})$	Thm 2.4	1st : $H^1(Y, T_Y)$	1st : T^1	Cor 5.2
Obst	: $H^1(Y, N_{Y/X})$	Thm 6.2	Obst : $H^2(Y, T_Y)$	Obst : T^2	Thm 10.2
(pro)-represent	: Hilb				

taking open cover

Fact (Cor 4.8)

non-sing affine sch have no non-triv deformation

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(10) Rem 1 : ① $\xrightarrow[\text{forget } X]{}$ ②

Y : non-singular $\subseteq \mathbb{P}^n$

$$0 \to T_Y \to T_{\mathbb{P}^n}|_Y \to N_{Y/\mathbb{P}^n} \to 0$$

$\overset{\text{long e.s.}}{\Longrightarrow}$ $\delta^0 : H^0(N_Y) \longrightarrow H^1(T_Y)$ boundary map
$\delta^1 : H^1(N_Y) \longrightarrow H^2(T_Y)$

See Prop 20.2.

(11) Rem 2 : $H^2(X, T_X) \neq 0$ is NOT necessary condition of unobstraction.

Thm (Bogomolov - Tian - Todolov)

X : weak C-Y s.t $H^0(X, T_X) = 0$.
then $\mathrm{Def}_X : \mathrm{Art}_k \to \mathrm{Sets}$
is formally smooth
Indeed pro-represented by $k[[X_1, \cdots, X_n]]$ $n = \dim_k H^1(X, T_X)$

Note • $H^0(X, T_X) = 0$ is natual
since for $n \geq 2$

$$H^0(X, T_X) \underset{\text{Serre}}{=} H^n(X, \Omega_X^1)^\vee \underset{\text{Hodge theory}}{``="} H^1(X, \omega_X^n) \underset{\text{C-Y}}{=} H^1(X, \mathcal{O}_X) = 0$$

• This thm suggest $\exists$ CY w/ $H^2(X, T_X) \neq 0$
but still the versal def is smooth

(12) Rem 3 : Another example
for line bundle, $T_2 \neq 0$ in general
but if char $k = 0$, $\mathrm{Pic}(X)$ representable
$\Rightarrow$ smooth (since it's group sch).

char $k = p \Rightarrow \mathrm{Pic}^0(X)$ not always smooth

☆ Comparison of Embedded and Abstract deformation

See §20.

X : non-sing surf of deg $d \geq 2$ in $\mathbb{P}^3$

using Euler seq

$\Rightarrow 0 \to \mathcal{O}_X \to \mathcal{O}_X(1)^4 \to T_{\mathbb{P}^3}|_X \to 0$

& $H^1(N_X) = H^1(\mathcal{O}_X(d)) = 0$

we get the following (see Eg 20.2.2)

d	$h^0(T_X)$	$h^0(T_{\mathbb{P}^3}\|_X)$	$h^0(N_X)$	$h^1(T_X)$	$h^1(T_{\mathbb{P}^3}\|_X)$
2	6	15	9	0	0
3	0	15	19	4	0
4	0	15	34	20	1
≥ 5	0	15	large	large	0
		↑ dim $PGL(3)$	↑ $\binom{d+3}{d}-1$	↑ use exact seq	↑ by Euler seq.

For example $d=2$; X: quadric

- No abstract deformation (= "local" rigid)
- $X \simeq \mathbb{P}^1 \times \mathbb{P}^1$: image of Segre emb. $\mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^3$
  $\Rightarrow$ it has 6-dim family of automorphism.
- Emb def has 9-dim family
  $\Rightarrow$ any two related by an automorphism of $\mathbb{P}^3$.

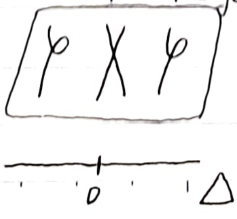
Note that $F_0 = \mathbb{P}^1 \times \mathbb{P}^1$ is "local" rigid but NOT "global" rigid

since $\exists$ family $f: \mathcal{X} \to \Delta$ s.t $\begin{cases} \mathcal{X}_s \simeq X & s \neq 0 \\ \mathcal{X}_0 \neq X \end{cases}$

Indeed, $\exists$

central

F_2 F_0 F_2 general



④ Deformation of Quotient

X: sch over k.

$q_0 : \mathcal{E}_0 \twoheadrightarrow \mathcal{F}_0$: quotient of (coherent sheaf) loc. free

w/ $Q_0 := \ker(q_0)$: loc. free

Note In order to have this property we will assume F_0 has the homological dimension $hd(F_0) \le 1$

See [Har2] ex III 6.5

$hd(F)$ = the least length of local free resolution of F

a) F is loc free $\Leftrightarrow hd(F) = 0$

b) $hd(F) \le n \Leftrightarrow ext^p(F, G) = 0$ $\forall p > n, \forall G \in Mod(X)$

Thm 7.2

Aut	Nothing
1st	$H^0(X, \mathcal{H}om(Q_0, \mathcal{F}_0))$
obst	$H^1(X, \mathcal{H}om(Q_0, \mathcal{F}_0))$

Note $E_0 = \mathcal{O}_{X_0} \Rightarrow F_0$: str sheaf of closed subsch $Y_0 \subseteq X_0$

then

$\mathcal{H}om(Q_0, \mathcal{F}_0) = Hom(I_{Y_0}, \mathcal{O}_{Y_0})$

$= N_{Y_0/X_0} \rightarrow$ ①

⑤ Coherent sheaf / loc. free

X: sch over k

F_0 : coh sheaf / loc. free

deformation (F, i) F : flat

$\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_{X_0} \overset{i}{\simeq} \mathcal{F}_0$ isom

X : deform of X_0.

Best situation = "$H^0(\mathcal{E}nd(E)) = k$"
= simple sheaf

This is motivation why we consider stable sheaves.

	coh. sheaf / loc free
Aut	$Ext^0(F) = H^0(X, \mathcal{E}nd(E))$
1st	$Ext^1(E) = H^1(X, \mathcal{E}nd(E))$
Obst	$Ext^2(E) = H^2(X, \mathcal{E}nd(E))$

In particular

⑥ Invertible sheaf

Thm 6.4

$Ext^i(F) = H^i(X, \mathcal{E}nd(F))$

$= H^i(X, \mathcal{O}_X)$.

why we need i? See [Har 1] 19.3.2 or Mukai's moduli book Ex 11.32

$E = \mathcal{O}(-1) \oplus \mathcal{O}(1)$ on $\mathbb{P}^1$

$H^1(\mathcal{E}nd(E)) = H^1(\mathcal{O}(-2)) \cong k$

& $H^2(\mathcal{E}nd(E)) = 0$ formally smooth.

However if we don't fix the isom with central fiber

$\forall \alpha \in Aut(E)$ bring a 1st def. into isom one.

Indeed $\exists\, \mathbb{C}^* \subset Aut(E)$ acting on $H^1(\mathcal{E}nd(E))$ non-trivially

(more precisely $\lambda \longmapsto \begin{pmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{pmatrix}$)

$\Rightarrow Def_E(k[\varepsilon]/(\varepsilon^2)) = \mathbb{C}/\mathbb{C}^*$

central

$\mathcal{O} \oplus \mathcal{O}$ $\mathcal{O}(1) \oplus \mathcal{O}(-1)$

general

This suggests

0 (ε)

jumping phenomena.

Other references (☆ : Best Recommeded).

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